

-- 1. TABLE CATEGORIE_ENTRETIEN (MaintenanceCategory)

```
CREATE TABLE MaintenanceCategory (  
  Id INTEGER PRIMARY KEY AUTOINCREMENT,  
  Name VARCHAR(50) NOT NULL UNIQUE,  
  Description TEXT,  
  DefaultCost REAL,  
  RecommendedInterval INTEGER, -- en km  
  CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

-- Données initiales

```
INSERT INTO MaintenanceCategory (Name, Description, DefaultCost,  
RecommendedInterval) VALUES  
(  
'Vidange', 'Vidange moteur + filtre à huile', 89.99, 15000),  
(  
'Freins', 'Changement plaquettes/disques', 150.00, 30000),  
(  
'Pneus', 'Changement de pneus', 300.00, 40000),  
(  
'Révision', 'Révision complète du véhicule', 200.00, 20000),  
(  
'Courroie', 'Changement courroie de distribution', 350.00, 80000),  
(  
'Batterie', 'Remplacement de la batterie', 120.00, 40000),  
(  
'Amortisseurs', 'Remplacement des amortisseurs', 250.00, 60000);
```

-- 2. TABLE UTILISATEUR

```
CREATE TABLE User (  
  Id INTEGER PRIMARY KEY AUTOINCREMENT,  
  Username VARCHAR(50) NOT NULL UNIQUE,  
  Email VARCHAR(100) NOT NULL UNIQUE,  
  Password VARCHAR(255) NOT NULL,  
  Role VARCHAR(20) NOT NULL DEFAULT 'Chef',  
  FirstName VARCHAR(50),  
  LastName VARCHAR(50),  
  Phone VARCHAR(20),  
  Address VARCHAR(255),  
  City VARCHAR(50),  
  PostalCode VARCHAR(10),  
  PhotoUrl VARCHAR(255),  
  CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  UpdatedAt DATETIME
```

);

-- 3. TABLE VEHICULE

```
CREATE TABLE Vehicle (  
    Id INTEGER PRIMARY KEY AUTOINCREMENT,  
    LicensePlate VARCHAR(20) NOT NULL UNIQUE,  
    Brand VARCHAR(50) NOT NULL,  
    Model VARCHAR(50) NOT NULL,  
    Year INTEGER,  
    Color VARCHAR(30),  
    CurrentMileage REAL DEFAULT 0,  
    FuelType VARCHAR(30),  
    FuelTankCapacity REAL DEFAULT 50,  
    Status VARCHAR(20) DEFAULT 'Disponible',  
    UserId INTEGER,  
    CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    UpdatedAt DATETIME,  
    FOREIGN KEY (UserId) REFERENCES User(Id) ON DELETE SET NULL  
);
```

-- 4. TABLE FUEL_RECORD

```
CREATE TABLE FuelRecord (  
    Id INTEGER PRIMARY KEY AUTOINCREMENT,  
    VehicleId INTEGER NOT NULL,  
    Date DATETIME NOT NULL,  
    Liters REAL NOT NULL,  
    Cost REAL NOT NULL,  
    Mileage REAL NOT NULL,  
    Station VARCHAR(50),  
    FuelType VARCHAR(30),  
    Driver VARCHAR(50),  
    Notes TEXT,  
    CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (VehicleId) REFERENCES Vehicle(Id) ON DELETE CASCADE  
);
```

-- 5. TABLE MAINTENANCE

```

CREATE TABLE Maintenance (
    Id INTEGER PRIMARY KEY AUTOINCREMENT,
    VehicleId INTEGER NOT NULL,
    Date DATETIME NOT NULL,
    Type VARCHAR(50) NOT NULL,
    Description TEXT,
    Cost REAL,
    MileageAtService REAL,
    Garage VARCHAR(100),
    InvoiceNumber VARCHAR(50),
    NextServiceDate DATETIME,
    NextServiceMileage REAL,
    Notes TEXT,
    CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (VehicleId) REFERENCES Vehicle(Id) ON DELETE CASCADE
);

```

-- 6. TABLE VEHICLE_SHEET (Fiche synthétique)

```

CREATE TABLE VehicleSheet (
    Id INTEGER PRIMARY KEY AUTOINCREMENT,
    VehicleId INTEGER NOT NULL UNIQUE,
    TotalDistance REAL DEFAULT 0,
    AvgConsumption REAL DEFAULT 0,
    LastMaintenance DATETIME,
    LastFuelDate DATETIME,
    TotalFuelCost REAL DEFAULT 0,
    MaintenanceCost REAL DEFAULT 0,
    CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    UpdatedAt DATETIME,
    FOREIGN KEY (VehicleId) REFERENCES Vehicle(Id) ON DELETE CASCADE
);

```

-- 7. TABLE STATISTICS

```

CREATE TABLE Statistics (
    Id INTEGER PRIMARY KEY AUTOINCREMENT,
    VehicleId INTEGER,
    Year INTEGER NOT NULL,

```

Month INTEGER NOT NULL,
TotalCost REAL DEFAULT 0,
TotalLiters REAL DEFAULT 0,
AvgConsumption REAL DEFAULT 0,
TotalMileage REAL DEFAULT 0,
RecordCount INTEGER DEFAULT 0,
CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (VehicleId) REFERENCES Vehicle(Id) ON DELETE CASCADE,
CONSTRAINT UQ_Statistics_YearMonth UNIQUE (VehicleId, Year, Month)
);